Tung Vo

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EDUCATION

University of St. Thomas - Houston, Texas *Expected May 2028*
Bachelor of Science in Applied Mathematics; Minor: Data Analytics - GPA: 3.91 / 4.00

Lone Star College - Houston, Texas *May 2025*
Honors Associate of Science, Summa Cum Laude - Distinguished Global Scholar (International Study)

PUBLICATIONS

Research Articles

(1) *Cayley Permutations, Fubini Rankings, and Unit Interval Parking Functions*
E. Barret, M. Beerbower, J. Elder, P. E. Harris, M. Sanchez, **T. Vo**, and C. Young. In preparation.

RESEARCH EXPERIENCE

Undergraduate Researcher, Rice University (CMOR Department) - Houston, TX *Fall 2026 - Present*
Research Advisor: Dr. Illya Hicks · Project Lead: Saber Dinpazhouh

- Use integer programming as a basis for developing an improved qubit routing optimal algorithm. Mapping logical qubits onto hardware-constrained physical qubits while minimizing the SWAP gates inserted during compilation.
- Benchmarking the heuristic against established routing methods (SABRE, TKET) on gate count and circuit depth to evaluate compilation efficiency.
- Scaling the algorithm to circuits of 20 qubits, where an exact integer-programming solvers became intractable and fail to return solutions within practical time limits at 14 qubits before.

Undergraduate Researcher, University of St. Thomas - Houston, TX *Fall 2025 - Present*
Research Advisor: Dr. Mary Flagg

- Defining and analyzing competitive zero forcing, a two-player graph coloring game that extends the classical zero-forcing color-change rule to an adversarial setting.
- Conducting theoretical analysis and computational modeling of vertex connectivity and forcing dynamics across multiple graph families.
- Presenting joint findings at the University of St. Thomas Annual Research Symposium (2026).

Undergraduate Researcher, University of Puerto Rico - Ponce - Ponce, PR *Summer 2026*
Research Advisor: Dr. Pamela Harris and Dr. Fernando Piñero

- Proving enumeration and structural results on weakly increasing and decreasing Fubini rankings, unit interval parking functions, and Cayley permutations under Dr. Pamela Harris.
- Characterizing the intersections of these three combinatorial families, including an if-and-only-if structural classification of $C_n \cap UPF_n$.
- Contributed to the mathematics and content of the research manuscript.

Undergraduate Researcher, Lone Star College Honors College - Houston, TX *Fall 2024*
Research Advisor: Dr. Kelly Phillips

- Examined how U.S. lottery-draft and Viet Cong propaganda recruitment tactics shaped soldier desertion rates (1950-1975).
- Analyzed primary-source propaganda posters and draft records, finding race and education level significantly affected draft likelihood.

Undergraduate Researcher, Lone Star College Honors College - Houston, TX *Spring 2024*
Research Advisors: Dr. Sean Tiffee, Dr. Rebecca Howard, Dr. Peggy Lambert, Prof. Casey Garcia

- Authored an autoethnography of competing at National Model UN-New York, applying triangulation and content analysis to examine how intercultural communication and public speaking drive leadership in large delegate settings.

Undergraduate Researcher, Lone Star College Honors College - Houston, TX *Spring 2024*
Research Advisor: Dr. Dana Van De Walker

- Applied a Most Similar System Design (MSSD) case-study framework to compare air pollution and life expectancy in the U.S. and Chad.
- Identified socioeconomic status, infant mortality, healthcare access, and reforestation investment as confounding drivers beyond pollution alone.

PRESENTATIONS

Vo, T. *"Competitive Zero Forcing on Paths and Cycles."* Texas Undergraduate Mathematics Conference, Stephen F. Austin State University, Nacogdoches, TX, 2026.

Vo, T., Barrett, E., & Sanchez, M. *"Enumerating Weak Order Unit Fubini Rankings"* Young Mathematician Conference, Ohio State University, Columbus, OH, 2026.

Vo, T., Cotsoradis, S., & Rutherford, C. *"Competitive Zero Forcing: A Novel Two-Player Graph Coloring Game."* Annual Research Symposium, University of St. Thomas, Houston, TX, 2026.

Vo, T. *"A Systemic Approach to Understanding the Natural World."* Rice University Environmental Data Academy Poster Session, Houston, TX, 2025.

Vo, T. *"A Systemic Approach to Understanding the Natural World."* Spring Honors & International Education Conference, Houston, TX, 2025.

Vo, T. *"The Fortunes of War or the Luck of the Draw: Recruitment Tactics and Desertion in the Vietnam War."* World History Association of Texas Annual Conference, Commerce, TX, 2025.

Vo, T. *"The Fortunes of War or the Luck of the Draw: Recruitment Tactics and Desertion in the Vietnam War."* Fall Honors & International Education Conference, Houston, TX, 2024.

Vo, T. *"The Fortunes of War or the Luck of the Draw: Recruitment Tactics and Desertion in the Vietnam War."* International Education Biennial Conference, Houston, TX, 2024.

Vo, T. *"Life Expectancy and Air Pollution: A Comparative Analysis of the United States and Chad."* Spring Honors & International Education Conference, Houston, TX, 2024.

OUTREACH

Finalist, CASE Program - Amideast Education Abroad Connect - Tunis, Tunisia *Summer 2025*

- Selected for an 8-day civil-society program engaging regional leaders on civic engagement, sustainability, and global justice.

Campus Representative, LSC Honors College Student Advisory Board *2024 - 2025*

- Advised the Associate Vice Chancellor on Honors College programming and system-wide student activities.

System Liaison Emissary, LSC Honors & International Education *2024 - 2025*

- Promoted Honors College, Rice Take Flight, Model UN, and scholarship programs; mentored the incoming liaison as Emeritus.

Press Team Member & Statesman, Texas Boys State (American Legion) *Summer 2022*

- Selected as 1 of 2 students from a class of 800; covered mock House proceedings and ran social-media campaigns.

AWARDS & HONORS

Dean's List, University of St. Thomas (Fall 2025, Spring 2026)	*2025 - 2026*
Hispanic-Serving Institution STEM (Dunn) Endowed Scholarship, University of St. Thomas	*2026*
Monaghan Excellence Scholarship, University of St. Thomas	*2025 - 2027*
Distinguished Global Scholar Study Abroad Scholarship, Lone Star College	*2025*
President's List, Lone Star College (All semesters)	*2023 - 2025*
Global Scholar Award & Language Scholarship, Lone Star College	*2023 - 2025*
Best in Committee & Outstanding Delegation, National Model UN, New York	*2024*
National Speech & Debate Association Academic All-American	*2022*

SKILLS

Environmental Programming & Tools: Python, R, spreadsheets, integer programming, LaTeX.
Languages: Vietnamese (native speaker), English (fluent)